

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 11
 - More on metric spaces
 - Metric subspaces

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Open sets

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The set A is **open** if every point in A is an interior point, or equivalently $\text{int}(A) = A$.

Challenge!

Problem

Consider the metric space $(\mathbb{R}, d_{\text{disc}})$ where d_{disc} is the discrete metric.

What sets are open sets?

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A point $x \in M$ is an **accumulation point** if for all $r > 0$ the ball $B_M(x; r)$ contains an element of A different from x .

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A point $x \in M$ is an **accumulation point** if for all $r > 0$ the ball $B_M(x; r)$ contains an element of A different from x .

The set $\bar{A}$ of all adherent points of A is called the **closure** of A .

Challenge!

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Which sets are closed?

Challenge!

Problem

Prove the following (Apostol Theorem 3.36). If (M, d) is a metric space, $U \subseteq M$ is open, and $C \subseteq M$ is closed, then $U \setminus C$ is open and $C \setminus U$ is closed.

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Theorem (Apostol Theorem 3.37)

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- (a) *A is closed*
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- (c) *A contains all of its accumulation points*
- (d) *$A = \bar{A}$*

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Let (M, d) be a metric space and $\{U_i : i \in I\}$ be a family of open sets in M . The union $\bigcup_{i \in I} U_i$ is open.

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Identical to the proof for $\mathbb{R}^n$. □

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Open balls in S :

$$B_S(x; r) = B_M(x; r) \cap S.$$

Example

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Open sets in subspaces

Theorem

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Furthermore $U \cap S = V$.



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We need to show that $V := U \cap S$ is open.

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Since $x \in V$ was arbitrary, this shows that V is open in S . □

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- $(0, 1/2)$ is open in both S and in M
- $(1/2, 1]$ is open in S , but not in M